# Solutions and teaching notes for chapter 6

**Teacher use.** The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. 6. *Action:* Revisit the rate $2a + h$.
2. $\mathbb{R} \setminus \{2\}$. *Action:* Set the denominator to zero.
3. Positive outside $[-2, 1]$, negative between the roots, zero at the endpoints. *Action:* Build two sign lines.
4. $x^3 - 2x^2 + x - 2$. *Action:* Distribute each term without omission.
5. They are equal. *Action:* Substitute -x in brackets.
6. $y = 2$. *Action:* Revisit the tangent equation and simplify.

## Investigation activity

$x^3$ is strictly increasing with a horizontal tangent at 0. For a differentiable increasing function, the rates on either side are positive or zero, so the derivative cannot be negative.

## Immediate applications

1. $5x^4$ on $\mathbb{R}$; $-\dfrac{2}{x^3}$ for $x \neq 0$; $\dfrac{1}{2\sqrt{x}}$ for $x > 0$.
2. $2x(3x-2) + 3(x^2+1) = 9x^2 - 4x + 3$; $-\dfrac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.
3. $f' = 2x - 4$, so decreasing up to 2 then increasing. Minimum $-3$ at 2; values at the endpoints 1 and $-2$, so maximum 1 at 0.
4. First even, second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equality at 1.
5. $A(x) = x(6-x)$ on $[0, 6]$; $A' = 6 - 2x$. Maximum 9 m$^2$ at 3.

## Exercises: method and results

**Exercise 6.1 – A collection of derivatives.** $12x^3 - 2$; $2x(x-1) + (x^2+2) = 3x^2 - 2x + 2$; $\dfrac{[2(x+3) - (2x+1)]}{(x+3)^2} = \dfrac{5}{(x+3)^2}$ for $x \neq -3$.

**Exercise 6.2 – A complete table.** $f' = 3(x-1)(x-3)$; signs $+, -, +$. Values $f(0) = 1$, $f(1) = 5$, $f(3) = 1$, $f(4) = 5$. Minimum 1 at 0 and 3; maximum 5 at 1 and 4. The endpoints count.

**Exercise 6.3 – A quotient.** $f' = 1 - \dfrac{4}{x^2} = \dfrac{(x-2)(x+2)}{x^2}$. Sign negative before 2, positive after. Minimum 4 at 2; for $x > 0$, $x + \dfrac{4}{x} \geq 4$, equality only at 2.

**Exercise 6.4 – Comparing two curves.** $f - g = x(x-1)(x+1)$. Signs: negative on $(-\infty,-1)$, positive on $(-1,0)$, negative on $(0,1)$, positive on $(1,+\infty)$. Intersection at $-1,0,1$. We can make the table of the three factors without differentiating.

**Exercise 6.5 – A box.** $V = x(12-2x)^2 = 4x(6-x)^2$. $V\prime = 12(x-2)(x-6)$, positive on $[0,2)$, negative on $(2,6)$. $V(0) = V(6) = 0$, $V(2) = 128$ cm$^3$. Maximum for x=2 cm; box $8 \times 8 \times 2$.

**Exercise 6.6 – Stationary without an extremum.** $f\prime = 3x^2$, so value 0 at 0. In every neighbourhood of 0, negative x give $f(x) < 0$ and positive x give $f(x) > 0$; no extremum.

**Exercise 6.7 – Comparing strategies.** Square: $(x-3)^2 + 3 \geq 3$. Derivative: $2x - 6$, minimum at 3, value 3. The first avoids a complete study of variation; the second illustrates the link between chapters.

**Exercise 6.8 – Delayed review.** It decreases on $(-\infty,0)$ and $(0,+\infty)$. It does not decrease on their union: $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$. The domain is not an interval.

**CQFD, exercise 12 p. 156, questions 1 and 2.** $f(x) - g(x) = \frac{1}{2}x^2 - 1 = \frac{1}{2}(x-\sqrt{2})(x+\sqrt{2})$. The curve of f is strictly above that of g for $x < -\sqrt{2}$ or $x > \sqrt{2}$, below between these two numbers; equality at the x-coordinates $\pm\sqrt{2}$.

**CQFD, exercise 16 p. 156, questions f and g.** For $f(x) = x^3$, we can choose $(-\infty,0)$: $f(x) < 0$ and the function is strictly increasing. For $g(x) = x^2 - 4$, we can choose $[0,2)$: $g(x) < 0$ and the function is strictly increasing on this interval. Subintervals that are not reduced to a point also work.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Integer power.
2. A. Rate of the reciprocal.
3. D. Product rule.
4. C. Sign of the derivative and variation.
5. B. Nonzero denominator.

**MCQ 2.**

1. C. $f(-x) = -f(x)$.
2. A. The other conclusions require more.
3. D. Formula and domain.
4. B. Reached at 1.
5. C. $A\prime(x) = 6 - 2x$ vanishes at 3, changing from positive to negative.

## True or false

1. False: $f(x) = \dfrac{1}{x} > 0$ on $(0, +\infty)$, but $f\prime(x) < 0$.
2. True, with the hypotheses stated; at the endpoints, the conclusion is not imposed.
3. False without sign hypotheses: on $[-2, -1]$, $u(x) = v(x) = x$ are increasing but $x^2$ is decreasing.

**Differentiated teaching.** For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.